

Assignment - 2

Code :

Banking System :

CREATE TABLE customer (

customer_id INT PRIMARY KEY,

customer_name VARCHAR(50) NOT NULL,

email VARCHAR(100) UNIQUE,

phone VARCHAR(15) UNIQUE NOT NULL,

city VARCHAR(30)

);

CREATE TABLE account (

account_id INT PRIMARY KEY,

account_type VARCHAR(20) NOT NULL,

balance DECIMAL(12,2) CHECK (balance >= 0),

customer_id INT NOT NULL,

CONSTRAINT fk_customer

FOREIGN KEY (customer_id)

REFERENCES customer(customer_id)

);

CREATE TABLE bank_transaction (

transaction_id INT PRIMARY KEY,

transaction_type VARCHAR(20) CHECK (transaction_type IN ('DEPOSIT','WITHDRAW')),

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amount DECIMAL(10,2) CHECK (amount > 0),
transaction_date DATE NOT NULL,
account_id INT,

CONSTRAINT fk_account
FOREIGN KEY (account_id)
REFERENCES account(account_id)
);

INSERT INTO customer (customer_id, customer_name, email, phone, city) VALUES
(1, 'Raj Mhase', 'raj@gmail.com', '9876543210', 'Pune'),
(2, 'Anita Sharma', 'anita@gmail.com', '9123456780', 'Mumbai'),
(3, 'Amit Patil', 'amit@gmail.com', '9988776655', 'Nashik'),
(4, 'Sneha Kulkarni', 'sneha@gmail.com', '9090909090', 'Kolhapur'),
(5, 'Rahul Verma', 'rahul@gmail.com', '9012345678', 'Delhi'),
(6, 'Priya Deshmukh', 'priya@gmail.com', '8899776655', 'Nagpur'),
(7, 'Karan Mehta', 'karan@gmail.com', '8787878787', 'Ahmedabad'),
(8, 'Neha Joshi', 'neha@gmail.com', '8666555444', 'Indore'),
(9, 'Vikas Rao', 'vikas@gmail.com', '8555444333', 'Hyderabad'),
(10, 'Pooja Singh', 'pooja@gmail.com', '8444333222', 'Bhopal');

INSERT INTO account (account_id, account_type, balance, customer_id) VALUES
(101, 'Savings', 50000, 1),
(102, 'Current', 75000, 2),
(103, 'Savings', 62000, 3),
(104, 'Savings', 48000, 4),
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(105, 'Current', 90000, 5),
(106, 'Savings', 55000, 6),
(107, 'Savings', 70000, 7),
(108, 'Current', 85000, 8),
(109, 'Savings', 60000, 9),
(110, 'Savings', 65000, 10);

INSERT INTO bank_transaction

(transaction_id, transaction_type, amount, transaction_date, account_id) VALUES

(1001, 'DEPOSIT', 10000, '2025-01-01', 101),
(1002, 'WITHDRAW', 5000, '2025-01-03', 102),
(1003, 'DEPOSIT', 15000, '2025-01-05', 103),
(1004, 'WITHDRAW', 3000, '2025-01-06', 104),
(1005, 'DEPOSIT', 20000, '2025-01-07', 105),
(1006, 'WITHDRAW', 7000, '2025-01-08', 106),
(1007, 'DEPOSIT', 12000, '2025-01-09', 107),
(1008, 'WITHDRAW', 6000, '2025-01-10', 108),
(1009, 'DEPOSIT', 18000, '2025-01-11', 109),
(1010, 'WITHDRAW', 4000, '2025-01-12', 110);

ALTER TABLE customer

ADD aadhaar_no VARCHAR(12) UNIQUE;

ALTER TABLE customer

MODIFY customer_name VARCHAR(100);

RENAME TABLE bank_transaction TO transactions;

SELECT * FROM customer;

SELECT * FROM account;

SELECT * FROM transactions;